We may calculate the periodic annuity cash flow (A), which occurs at the end of each respective period, as follows:

$$A = \frac{r(PV)}{1 - (1 + r)^{-t}}, \quad (8)$$

where:

A = periodic cash flow,

r = market interest rate per period,

PV = present value or principal amount of loan or bond, and

t = number of payment periods.

EXAMPLE 5

Mortgage Cash Flows

An investor seeks a fixed-rate 30-year mortgage loan to finance 80 percent of the purchase price of USD1,000,000 for a residential building.

1. Calculate the investor's monthly payment if the annual mortgage rate is 5.25 percent.

Solution:

Solve for A using Equation 8 with r = 0.4375% (= 5.25%/12), t of 360, and PV of USD800,000 (=80% × USD1,000,000):

$$A = USD4{,}417.63 = \frac{0.4375\%\,(\text{USD}800{,}000)}{1 - (1 + 0.4375\%)^{-360}}.$$

The 360 level monthly payments consist of both principal and interest.

2. What is the principal amortization and interest breakdown of the first two monthly cash flows?

Solution:

Month 1 interest is USD3,500 and principal is USD917.63
Month 2 interest is USD3,495.99 and principal is USD921.64

Month 1:

Interest: USD3,500 = USD800,000 × 0.4375% (=$PV_0 \times r$)

Principal Amortization: USD4,417.63 – USD3,500 = USD917.63

Remaining Principal (PV_1): USD799,082.37 = USD800,000 – USD917.63

Month 2:

Interest: USD3,495.99 = USD799,082.37 × 0.4375% (=$PV_1 \times r$)

Principal Amortization: USD4,417.63 – USD3,495.99 = USD921.64

Remaining Principal (PV_2): USD798,160.73 = USD799,082.37 – USD921.64

Although the periodic mortgage payment is constant, the proportion of interest per payment declines while the principal amortization rises over